

Sairam Bokka

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SKILLS

Languages: Python, TypeScript, JavaScript, HTML5, CSS

Frontend: React, Next.js, Tailwind CSS, WebAssembly (Pyodide), Context API, React Hooks

Backend: Node.js, Express.js, FastAPI, Spring Boot, RESTful APIs, WebSocket, Socket.IO

Infrastructure & Tools: Docker, Docker Compose, Jenkins, Git, CI/CD, Linux, Playwright, Jest

AI & Security: LangChain, CrewAI, PyTorch, Scapy, CompTIA Security+, Splunk, Nmap

WORK EXPERIENCE

Corca Research Inc

Remote – Baltimore, MD

Full stack Software Developer

Dec 2025 – Present

- Promoted from Engineering Intern to full-time after shipping core editor features; delivered 65+ merged pull requests (~20K lines across 450+ files) on a browser-based math editor, working test-first (TDD).
- Shipped a NumPy + Plotly data-analysis subsystem — users select arrays or matrices to generate 8+ interactive chart types (histogram, boxplot, scatter, FFT, polynomial fit) and descriptive statistics, entirely in-browser via Pyodide/WebAssembly.
- Owned the Phantom Hints (tab-to-autocomplete) engine end to end — multi-line variable context, unit carry-through, and hint ordering across 10+ math domains: statistics, matrix operations, set theory, inequalities, limits, and Taylor series.
- Extended the in-browser SymPy transpiler with 15+ new function classes (exotic/hyperbolic trig, 10 statistical measures, set theory, matrix det/inv/transpose) and designed a non-invasive unit-preservation pipeline across variables, constants, and the object panel.

RootsID LLC

Baltimore, MD

Software Engineering Intern

Feb 2025 – May 2025

- Refactored a JavaScript/TypeScript pilot codebase into 5+ reusable Node.js packages following OOP principles, reducing development time by 30% and eliminating 200+ lines of duplicate code.
- Orchestrated containerized deployment for 4+ microservices across Linux environments, reducing system setup latency by 25% through automated Docker Compose workflows; resolved 15+ critical issues, cutting crashes by 30%.
- Built a secure in-browser document-signing feature with RESTful APIs and KERI-based credentials, enabling 100+ users to process and sign documents.

Tenable Network Security

Baltimore, MD

Security Engineering Intern

June 2024 – August 2024

- Developed Python-based Slack bot with asynchronous RESTful API integration for automated lookups across 8 repositories, reducing response time from 3–5 days to under 24 hours and improving efficiency by 80%.
- Containerized application using Docker with comprehensive unit testing in Python, implementing 50+ tests, achieving 95% code coverage for production reliability.
- Built Jenkins-based CI/CD pipeline integrated with Git for automated builds, testing, and deployment, reducing manual effort by 40% and supporting scalable rollout of internal tools.

Concentrix Catalyst

Hyderabad, IN

Software Engineer

March 2022 – June 2023

- Contributed to the core architecture of 4 full-stack applications, delivering high-impact features across the complete product lifecycle.
- Collaborated with a 5-person cross-functional team to build a dynamic article landing page system in Adobe Experience Manager (AEM).
- Created modular AEM templates that empowered 8+ non-technical authors to launch high-traffic landing pages independently through dynamic JSON data retrieval.
- Optimized user experience by implementing asynchronous content loading via AJAX, ensuring seamless data rendering and faster page load speeds.

PROJECTS

- **PR Dashboard (TypeScript, GitHub API, Cloudflare Workers)** — Linear-style dashboard for open GitHub PRs with live polling, unread-delta badges, CI status, and browser notifications; OAuth via a single Cloudflare Worker with no server-side secret storage.
- **Financial Analyzer (Next.js, TypeScript, Claude AI)** — Full-stack platform using the Claude AI API to interpret financial data and deliver AI-generated insights through a responsive Next.js frontend; containerized with Docker.
- **Real-Time Scalable Chat Application (React, Socket.IO, Node.js)** — Messaging platform with bidirectional WebSocket communication and JWT + Google OAuth2 authentication.

EDUCATION

University of Maryland, Baltimore County

Baltimore, Maryland

Master of Science in Cybersecurity, GPA 4.0/4.0

May 2025

Guru Nanak Institutions Technical Campus

Hyderabad, India

Bachelor of Technology in Electronics and Communications Engineering, GPA 3.6/4.0

August 2022